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Diploma in Computer Science (DCS)

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MPU2332WA Green Future Malaysia

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Individual Assignment (10%)

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Recycling

Recycling is the process of turning waste into new products and materials. This concept frequently takes the recovery of energy from waste materials into account. The capacity of a material to regain the qualities it had in its initial state determines how recyclable it is. (G Villalba, M Segarra, A.I Fernández, J.M Chimenos, F Espiell, 2002) It is an alternate method of disposing of waste to "conventional" methods that can conserve resources and reduce greenhouse gas emissions. Additionally, by preventing the waste of potentially useful materials and lowering the consumption of fresh raw materials, it can lower energy consumption as well as air and water pollution from landfills and incineration. (Jens lienig, Hans Bruemmer, 2017) Recycle is the third component of the waste hierarchy which is reduce, reuse, and recycle. (Jens lienig, Hans Bruemmer, 2017) The key to modern waste reduction is recycling. By eliminating raw material input and rerouting waste output in the economic system, it fosters environmental sustainability. (Geissdoerfer, M., Savaget, P., Bocken, N., & Hultink, E., 2017).



Recycling bin in Japan



The red warning sticker

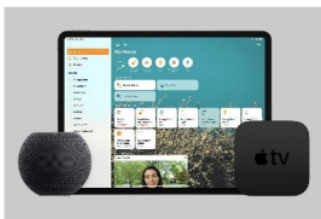


The sorting guide poster in Nagoya, Japan

Some nations, like Japan, are well known for their recycling laws. The Act on the Promotion of Sorted Collection and Recycling of Containers and Packaging has been in effect in Japan since April 1997, and it aims to reduce waste of glass containers, PET bottles, and paper cartons. Since April 2000, plastic packaging and containers other than PET bottles have been included. (Ministry of the Environment, Japan., 2005) The Japanese Container and Package Recycling Association (JCPRA), a government-designated organization that was founded on September 25, 1996, is in charge of recycling per the act. This Act mandates that consumers follow the municipally established sorting guidelines. The sorted waste is then collected by the municipalities and stored for collecting by the recycling company. Then, based on the volume of product they produce or sell, JCPRA will charge manufacturers and business entities that use containers and packaging a recycling fee. Every year, recycling business entities are chosen through a public bidding process in every local municipality where a waste storage site is located. The waste must be gathered and transported from the storage areas to the recycling facilities by them. To guarantee that the waste is recycled, these recycling business entities only receive payment after submitting a delivery report that has been signed by the recipient of the recycled goods. If something was disposed of improperly, a large red warning sticker is placed on the offending trash bag to shame the offender. (Wrigley, 2014)

Internet of Things (IoT)

The Internet of Things, or IoT, is a network of interrelated devices that connect and exchange data with other IoT devices and the cloud. (Gillis, 2023) By enabling everyday objects to gather, process, and share information frequently without human assistance, this idea fundamentally broadens their capabilities. Electronics, communication, and computer science engineering are all included in the internet of things. The wide range of IoT devices are typically divided into the infrastructure, infrastructure space, commercial space, and consumer space. (Charith Perera, Chi Harold Liu, Simal Jayawardena, 2015) IoT devices are frequently found at the consumer level in the idea of home automation. A platform or hubs that manage smart appliances and devices may serve as the foundation of a smart home or automated home. For instance, manufacturers can use Apple's HomeKit to control their home furnishings and accessories through an app on iOS devices like the iPhone or Apple Watch. It could be a particular app or a native iOS app like Siri. (Ben L, 2018) In commercial or Industrial, IoT devices can often saw in manufacturing industry since IoT can connect various manufacturing devices equipped with sensing, identification, processing, communication, actuation, and networking capabilities. Intelligent IoT systems allow for quick product development, optimization, and market demand response. For instance, most businesses in the instant food sector, like Maggie, will have sensors on their packaging equipment that will enable them to determine the food's quality and maintain a high standard for it. One of the primary applications for the IoT in the infrastructure sector is the monitoring and management of the operation of sustainable urban and rural infrastructures like bridges, railroad tracks, and on- and offshore wind farms. (Jayavardhana Gubbi, Rajkumar Buyya, Slaven Marusic, Marimuthu Palaniswami, 2013) Any changes to structural conditions or events that might jeopardize safety or raise the risk can be tracked using the IoT infrastructure. The IoT can improve productivity, increase efficiency, and improve the quality of workdays in the construction industry.



Apple HomeKit



Inductive Wheel Sensor

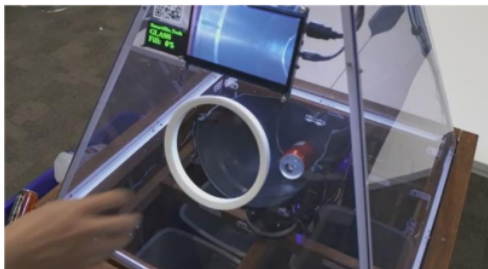


X-ray inspection system in food industry

A Green City in Penang

In my opinion, I think Penang can use the intelligent recycle bin to replace with the normal recycling bin in the state. Based on the analysis did by Reel paper, the reason that people do not recycle is they confuse about what to recycle and how, inconvenience and a lack of access to recycling, lack of space to accommodate recyclables and some of them are just simply forgetting to recycle. (Kinejara, 2022) So, by replace the normal recycle bin with intelligent recycle bin, people will more easy to do recycle. This is due to intelligent recycle bin will sorts the recyclables by itself using its sensor to sense the weight, matter and material. For example, when people throw a plastic bottle into it, the sensor will sense the weight and materials of the waste and determine whether it is recyclable or not. If it is recyclable, it will the recycle bin will automatically put the waste into the correct bin inside it. If not, it will just sort the waste as a non-recyclable waste and put it in the following bin.

I believe that using the intelligent recycle bin could help to create a sustainable city in Penang because this recycle bin will help to solve the problem that some of the citizens do not know what and how to recycle. All the thing that they need to do is just throw their waste in it and the bin will help them to sort. Furthermore, by using the intelligent recycle bin, the collector of these recycle waste do not need to use their time to sort again the waste since it's already sorted automatically when people throw their waste into it. All they need to do is just need to dispose those waste or send it to the processing factory. By using an intelligent recycle bin, it can also help to solve the problem which is lack of space to accommodate recycle bin. When we using normal recycle bin, we need to place 4 bins with specific colour for people to throw their waste and it is very occupying space. By using the intelligent recycle bin, we only require to place one bin to replace with 4 bins. It is space-saving and help to reduce the cost. Put this type of recycle bin on the street also can let others more prefer to throw their waste into it since it is new and intelligent. In Denmark, they also use the intelligent recycle bin to collect the waste. Based on the research, this can help to increase the recycling rate. (Making the World's Waste, n.d.)



Picture of the intelligent recycle bin develop by Dr Xu Wang, Lecturer of University of Technology Sydney

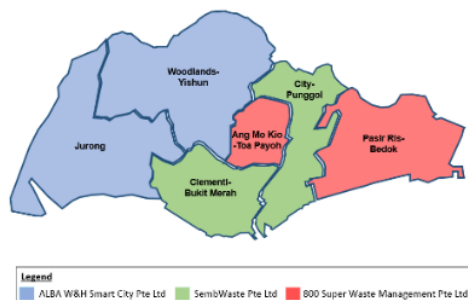


Picture of the intelligent recycle bin in Copenhagen, Denmark

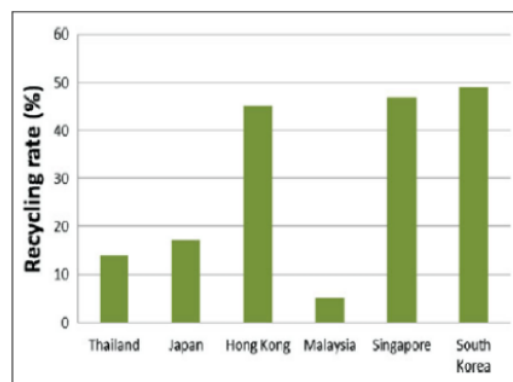
Malaysia vs Singapore

There are some differences between Malaysia and Singapore in Recycling. One of it was Government Support and Regulation. In Singapore, their government actively encourages IoT-enabled recycling through rules, rewards, and extensive waste management policies. For example, Singapore's National Recycling Program launched in April 2001. Under this program, recycling bins and recycling collection services must be offered by public waste collectors (PWCs) licensed by the National Environment Agency (NEA) in all HDB estates, private landed properties, condominiums, and private apartments enrolled in the public waste collection scheme. (National Environment Agency Singapore, 2022) In Malaysia, Malaysia government also have shown interest but different regions may have different levels of government backing and enforcement. For example, Malaysia government also have collab with a waste management company called SWM Environment to launch a program called KITARecycle but this programme only available for the citizen who residing in Negeri Sembilan, Melaka and Johor. In this programme, the citizen that stay in those area can download their application and register an account on it. After that, they can send their recyclable waste to their centre and the collector there will scan their QR code that paste on the bags and the user will receive the Rpoint. These Rpoint can be used to redeem money. (kitarcycle, n.d.)

Furthermore, there are also a difference in public awareness. Singapore places a lot of emphasis on educating the public about recycling and the Internet of Things' contribution to better waste management. For example, Singapore government have held the education campaign and always have spot check on the recycling bin at the household area cause their citizen more pay attention on recycling their waste (Tan, 2019). In Malaysia, our efforts to raise public awareness might not be as widespread or consistent. For example, although Malaysia have Recycle program launch by government, but very less people would follow it. This is due to most Malaysian are unaware of the high commodity value of the rubbish they generate and some of them are just simply forgetting to recycle. (Noor, n.d.)



Recycle Map of Singapore



Graph of Recycle rate

References

1. Ministry of the Environment, Japan. (2005). Law for the Promotion of Sorted Collection and Recycling of Containers and Packaging.
2. (2022, November 18). Retrieved from National Environment Agency Singapore: <https://www.nea.gov.sg/our-services/waste-management/3r-programmes-and-resources/national-recycling-programme>
3. Ben L. (2018, August 31). *HomeKit devices getting more affordable as Lenovo announces Smart Home Essentials line*. Retrieved from 9to5Mac: <https://9to5mac.com/2018/08/31/cheap-homekit-bulbs-switches-camera/>
4. Charith Perera, Chi Harold Liu, Srimal Jayawardena. (2015, January 31). *The Emerging Internet of Things Marketplace From an Industrial Perspective: A Survey*. Retrieved from arxiv: <https://arxiv.org/abs/1502.00134>
5. G Villalba, M Segarra, A.I Fernández, J.M Chimenos, F Espiell. (2002). A proposal for quantifying the recyclability of materials.
6. Geissdoerfer, M., Savaget, P., Bocken, N., & Hultink, E. (2017). The Circular Economy – A new sustainability paradigm? *Journal of Cleaner Production*, 757-768.
7. Gillis, A. S. (2023, August). *internet of things (IoT)*. Retrieved from TechTarget: <https://www.techtarget.com/iotagenda/definition/Internet-of-Things-IoT>
8. Jayavardhana Gubbi, Rajkumar Buyya, Slaven Marusic, Marimuthu Palaniswami. (2013). Internet of Things (IoT): A vision, architectural elements, and future directions. *Future Generation Computer Systems*, 1645-1660.
9. Jens Ienig, Hans Bruemmer. (2017). Recycling Requirements and Design for Environmental Compliance. *Fundamentals of Electronic Systems Design*, 193-218.
10. Kinejara, A. (2022). *Why Do People Not Recycle Anymore?*
11. *kitarcycle*. (n.d.). Retrieved from <https://kitarcycle.com/>
12. *Making the World's Waste*. (n.d.). Retrieved from Nordsense: <https://nordsense.com/>
13. Noor, Z. Z. (n.d.). *TOWARDS SUSTAINABLE HOUSEHOLD WASTE MANAGEMENT IN URBAN AREAS*. Retrieved from MIT-UTM: <https://malaysiacities.mit.edu/papernoor>
14. Tan, A. (2019). *Education campaigns and spot checks on recycling bins among ideas to improve household recycling*.
15. (2022). *This AI-powered 'smart bin' sorts recycling by itself*. Euronews.
16. Wrigley, F. (2014). "Recycling in Japan" or "Reasons to get it right and avoid eternal shame". SoraNews 24.

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